Modeling and Analyzing Interactions Between Stakeholders for Train Decarbonization Decisions at a Regional Scale

Benoît Volant[12], Franck Marle[1], Yann Leroy[1], Jeanne-Marie Dalbavie[2]

[1]Laboratoire Genie Industriel, CentraleSupélec, Université Paris-Saclay
[2]Ikos Consulting

Abstract: This study addresses the complex network process regarding the technological swap of diesel-powered regional trains to reach the Paris Agreement's targets. Interviews with technical experts from Authorities for transport Organization on their Territory (AOT) allowed to build a stakeholder influence model, studying the interaction dynamics within this network. Thanks to a Design Structure Matrix (DSM) approach, we analyzed the complex stakeholder network influencing decarbonization strategies at the regional level. We investigate the workflows of various actors, including public authorities, industry experts, and technology providers, in selecting traction technologies. Our DSM analysis reveals the dominant roles of key public entities and strategic industrial stakeholders and highlights the challenges of integrating multicriteria considerations into the decision-making process. This research contributes to the DSM application for decision support with actor-actor workflows. The proposed methodology offers a structured framework for decision-making, reducing private sector bias and aligning with regional and environmental imperatives.

Keywords: stakeholder network, decision process clusters, topological analysis, railway decarbonization

1 Introduction

As ecological transition and the Paris Agreement call for a gradual decarbonization of transportation, the rail sector must contribute by replacing current fossil fuels-based traction. The Paris Agreement roadmap calls for a 55% reduction in Greenhouse Gas (GHG) emissions by 2030 compared to 1990 levels (European Climate Law, 2021). Currently, 0.3% of CO_2 emissions result from using fossil fuels in railways for 8% of the world's motorized passenger movements (Mandegari et al., 2023; Rungskunroch et al., 2021). Therefore, a global and ambitious decarbonization approach must be considered to improve this performance in low-emission transportation. The French railway network comprises 65% of electrified lines, mostly on high-density lines. 85% of the 9 137 km of low-density lines are powered by diesel trains (Ministère de la transition écologique et solidaire, 2020). French territory is divided into regions, each owning its Authority for transport Organization on their Territory (AOT) that possesses its own trains. Regional trains undergo major maintenance when they reach half-life to ensure their 40-year lifetime. This operation is a favorable context for replacing their energy storage system. AGC, a highly represented model of French trains, will reach its half-life between 2024 and 2031.
However, there are no proven cost-efficient solutions today to replace diesel-powered trains (European Commission, 2017). Electrification was a promising alternative. However, this solution's infrastructural costs are too high to be acceptable in low-density lines. New alternatives are emerging, offering potentially more environmentally friendly and cost-efficient perspectives, such as:

- Battery-powered traction, heavily inspired by the automotive sector and recent developments in lithium batteries. The high cost of lithium-ion batteries is widely seen as the most significant barrier to mass adoption (European Commission, 2017).
- The use of various biofuel technologies to power a combustion engine (Bernard & Prieur, 2006).
- Using a hydrogen fuel cell with batteries to power an electric motor (Akhoundzadeh et al., 2021).
- These new technologies can also be hybridized (Meynerts et al., 2018).

Different technological maturities characterize these alternatives.

The problem is that, although promising, these technologies' maturities are low and thus difficult to assess in terms of various parameters, such as their potential for reducing environmental footprint, their performance (energy efficiency, autonomy, maintainability, lifespan) and their impact on the rest of the lifecycle (the production, the infrastructure, and the maintenance for instance). The target of reducing greenhouse gas emissions during the use phase generally results in environmental impacts shifting towards equipment production phases, infrastructure construction, and end-of-life management. Additionally, consideration of development, deployment, and operational costs is necessary. Uncertainty varies with technology maturity, and a choice of technology implies tradeoffs in the decision criteria. Reaching a decision on this technological swap (referred to as train decarbonization in the paper) is thus a methodological and practical challenge due to the number and diversity of decision criteria, alternatives, and involved actors. It is a crucial issue since closing low-density lines when diesel trains cannot circulate anymore has societal and economic consequences for the territory. Keeping only diesel is impossible; however, closing the lines is not desired either.

A specific decision process assisting Authorities for transport Organization on their Territory (AOT) must be created to consider the technical, environmental, societal, and economic targets and constraints. This process must be constructed in